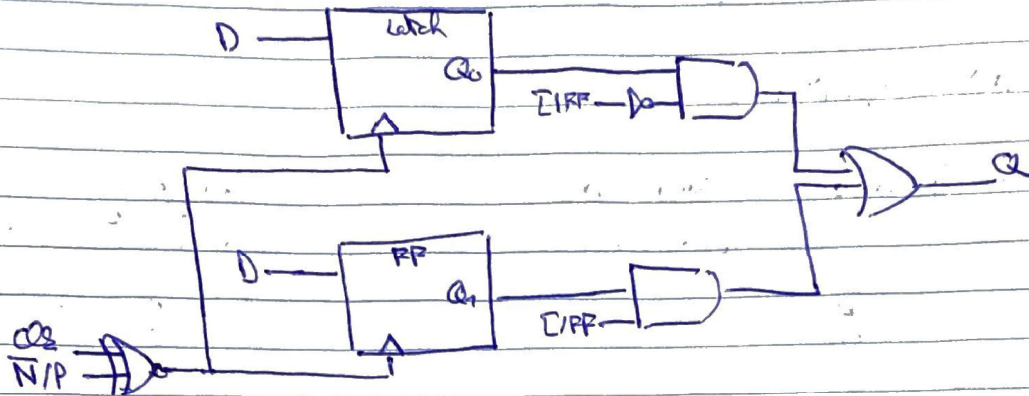
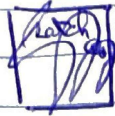


Hence, the final implementation is the following:

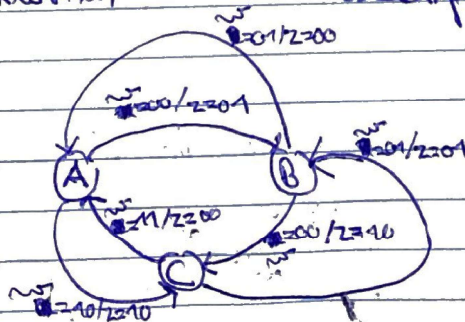


Question 5:

input: 00000000
output: 00000000

input	output
one employee goes in	0 0
one employee goes out	0 1
two employees go in	1 0
two employees go out	1 1

States:
A: no one in
B: one in
C: two in



input	output	z1	z0
no one in	0 0	0	0
one in	0 1	0	1
two in	1 0	1	0
two out	1 1	1	1